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Robust Nonlinear Predictive Control with Modeling Uncertainties and Unknown Disturbance for Single-Link Flexible Joint Robot

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***Abstract* - A robust nonlinear predictive controller for single-link flexible joint robot is presented in this paper. The objective is to track some predefined profiles for angular displacement of the link. The prediction model, used in the controller design, is carried out via Taylor series expansion. The uncertainties and error modeling of the system are taken into account by this controller. In order to deal with them, a disturbance observer is designed from the predictive control law. Simulation results show the high performance of the proposed control scheme.**

Index Terms – Flexible joint robot, predictive control.

I. INTRODUCTION

The flexible joint robot is modeled by a nonlinear representation to describe its dynamic behavior. However, this mathematical model is only an approximation of the real system. The simplified representation of the system behavior contains model inaccuracies such as parametric uncertainties, unmodeled dynamics, and external disturbances [9, 11]. Because of these inaccuracies, the controller design must take into account their effects in order to improve the performance of the closed loop system.

Model based predictive control (MPC) has received a great deal of attention and is considered by many to be one of the most promising methods in control engineering. The predictive control strategy belongs to the optimal control methods. The difference is that the cost function to be optimized is defined over a future horizon. In traditional optimal control law, the online computation burden is heavy to solve the optimization problem [1, 5, 11], which is unacceptable for systems characterized by fast dynamics like robotics. To overcome this advantage, several works, about nonlinear predictive control, have been done as in [2, 10], where the model output prediction for error tracking is obtained by expanding the output signal and the reference signal. Then, the optimization of the predictive tracking errors is used to derive the offline control laws. This method applied for mechanical systems with good performance [4, 6, 8]. In addition of that, to solve the problem of modeling inaccuracies, one of the methods is to design an observer to deal with these uncertainties. A nonlinear disturbance observer have been developed in [3], and applied in case of mechanical system with good performance for disturbance rejection [4].

The aim of this paper is to design a nonlinear predictive controller for the single-link flexible joint robot with a disturbance observer to deal with uncertainties and unmodeled quantities. It is organized as follows. Section II describes the mathematical model of the robot. Section III derives the nonlinear predictive controller in detail. The objective is to track the angular displacement of the link. Section IV presents the design of the disturbance observer to take into account the mismatched model and external disturbance. Section V gives the stability analysis of the closed-loop system. Section VI presents simulation results for tracking problem of the robot using the proposed controller. Section VII concludes the paper.

II. MATHEMATICAL MODEL OF THE ROBOT

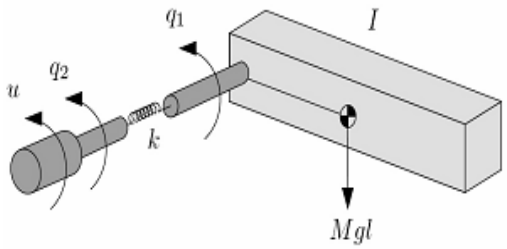


Fig. 1 Single-link flexible joint robot [12]

Consider the single-link flexible joint robot shown in Fig. 1. For simplicity, the viscous damping is neglected in system modeling. The equations of motion are [9, 12]

$$\begin{aligned} I\ddot{q}_1 + Mgl\sin(q_1) + k(q_1 - q_2) &= 0 \\ J\ddot{q}_2 + k(q_2 - q_1) &= u \end{aligned} \tag{1}$$

where, q_1 is the link angular displacement, q_2 is the motor angular position, I is the link inertial, J is the rotor inertia, k is the stiffness , M is the link mass, g is the gravity constant, and l is the center of mass. The control u is the torque delivered by the motor.

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In order to built the state space model, the angles and velocities, which are assumed to be known by measurement, are taken as state variables

$$x_1 = q_1; x_2 = \dot{q}_1; x_3 = q_2; x_4 = \dot{q}_2; \tag{2}$$

Then, the system (1) is written as

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{Mgl}{I}\sin(x_1) - \frac{k}{I}(x_1 - x_3) \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \frac{k}{J}(x_1 - x_3) + \frac{1}{J}u \end{aligned} \tag{3}$$

The state space model is given under the form

$$\dot{x} = f(x) + g_1(x)u \tag{4}$$

with

$$f(x) = \begin{bmatrix} x_2 \\ -\frac{Mgl}{I}\sin(x_1) - \frac{k}{I}(x_1 - x_3) \\ x_4 \\ \frac{k}{J}(x_1 - x_3) \end{bmatrix}; g_1(x) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{J} \end{bmatrix}$$

The output to be controlled is the link's angle

$$y = h(x) = x_1 \tag{5}$$

III. Nonlinear Predictive Control

The predictive control algorithm belongs to the optimal control. The difference is that the cost function is defined over a future horizon. It can be defined by the simple quadratic form

$$\Im = \frac{1}{2}\int_0^T e(t+\tau)^2 d\tau \tag{6}$$

$e(t+T) = y(t+T) - y_{ref}(t+T)$ is the tracking error at the next step $(t+T)$.

T is the prediction time, $y(t+T)$ a T-step ahead prediction of the system output and $y_{ref}(t+T)$ the future reference trajectory. The control weighting term is not included in the performance index (2); rather, a weighting can be achieved by the predictive time [2, 4].

The prediction output can be carried out from the Taylor series expansion

$$\begin{aligned} y(t+T) = h(x) + TL_f h(x) + \frac{T^2}{2!}L_f^2 h(x) + \ldots + \frac{T^r}{r!}L_f^r h(x) \\ + \frac{T^r}{r!}L_g L_f^{r-1} h(x)u(t) \end{aligned} \tag{7}$$

r is the relative degree defined to be the number of times of output differentiation until the control input appears.

The Lie derivative of h with respect to f, denoted $L_f h$, is defined as

$$L_f h = \frac{\partial h}{\partial x} f(x) = \sum_{i=1}^{n} \frac{\partial h}{\partial x_i} f_i(x) \tag{8}$$

Iteratively, we have

$L_f^k h = L_f(L_f^{k-1} h)$ for k=1,…,n;

and $L_{g_1} L_f h = \frac{\partial L_f h}{\partial x} g_1(x)$

Lie derivatives of the link angle output are

$$\begin{aligned} y &= h \\ \dot{y} &= L_f h \\ \ddot{y} &= L_f^2 h \\ \dddot{y} &= L_f^3 h \\ y^{(4)} &= L_f^4 h + L_{g_1} L_f^3 h u \end{aligned} \tag{9}$$

where,

$$\begin{aligned} h &= x_1 \\ L_f h &= x_2 \\ L_f^2 h &= -\frac{Mgl}{I}\sin(x_1) - \frac{k}{I}(x_1 - x_3) \\ L_f^3 h &= -\frac{Mgl}{I}\cos(x_1)x_2 - \frac{k}{I}(x_2 - x_4) \\ L_f^4 h &= \frac{Mgl}{I}\sin(x_1)\left(x_2^2 + \frac{Mgl}{I}\cos(x_2) + \frac{k}{I}\right) + \\ &\quad \frac{k}{I}(x_1 - x_3)\left(\frac{k}{I} + \frac{k}{J} + \frac{Mgl}{I}\cos(x_1)\right) \\ L_{g_1} L_f^3 h &= \frac{k}{IJ} \end{aligned} \tag{10}$$

The relative degree of the system is $r = 4$.

Then,

$$\begin{aligned} y(t+T) = h(x) + TL_f h(x) + \frac{T^2}{2!}L_f^2 h(x) + \frac{T^3}{3!}L_f^3 h(x) + \\ \frac{T^4}{4!}L_f^4 h(x) + \frac{T^4}{4!}L_g L_f^3 h(x)u(t) \end{aligned} \tag{11}$$

Similarly, the prediction of the reference may be expanded in Taylor series expansion

$$\begin{aligned} y_{ref}(t+T) = y_{ref}(t) + T\dot{y}_{ref}(t) + \frac{T^2}{2!}\ddot{y}_{ref}(t) + \frac{T^3}{3!}\dddot{y}_{ref}(t) + \\ \frac{T^4}{4!}y_{ref}^{(4)}(t) \end{aligned} \tag{12}$$

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Then, the predicted error is given by

$$\begin{aligned} e(t+T) &= y(t+T) - y_{ref}(t+T) \\ &= \mathrm{T}\left(\mathrm{Y}(t) - \mathrm{Y}_{\mathrm{r}}(t)\right) \end{aligned} \tag{13}$$

where,

$$\mathrm{T} = \left[1 \quad T \quad \frac{T^2}{2} \quad \frac{T^3}{6} \quad \frac{T^4}{24}\right];$$

$$\mathrm{Y}(t) = \begin{bmatrix} y(t) \\ \dot{y}(t) \\ \ddot{y}(t) \\ \dddot{y}(t) \\ y^{(4)}(t) \end{bmatrix} = \begin{bmatrix} h \\ L_f h \\ L_f^2 h \\ L_f^3 h \\ L_f^4 h \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ L_g L_f^3 hu(t) \end{bmatrix};$$

$$\mathbf{Y}_{\mathbf{r}}(t) = [y_{ref}(t) \quad \dot{y}_{ref}(t) \quad \ddot{y}_{ref}(t) \quad \dddot{y}_{ref}(t) \quad y_{ref}^{(4)}(t)]^T$$

The cost function (6) is rewritten as

$$\Im = \frac{1}{2}\left(\mathrm{Y}(t) - \mathrm{Y}_{\mathrm{r}}(t)\right)^T \Pi \left(\mathrm{Y}(t) - \mathrm{Y}_{\mathrm{r}}(t)\right) \tag{14}$$

where,

$$\Pi = \int_0^T \mathrm{T}^T \mathrm{T}\, d\tau$$

The optimal control is obtained by putting $\partial\Im/\partial u = 0$

$$u(t) = -(L_g L_f^3 h)^{-1}\left(\mathrm{KM}_4 + L_f^4 h - y_{ref}^{(4)}\right) \tag{15}$$

with

$$\mathrm{M}_4 = \begin{bmatrix} h - y_{ref} \\ L_f h - \dot{y}_{ref} \\ L_f^2 h - \ddot{y}_{ref} \\ L_f^3 h - \dddot{y}_{ref} \end{bmatrix}$$

The elements of K are carried out from the matrix Π

$$\begin{aligned} \mathrm{K} &= [k_1 \quad k_2 \quad k_3 \quad k_4] \\ &= \left[\frac{2557}{60*T^4} \quad \frac{2557}{72*T^3} \quad \frac{2557}{168*T^2} \quad \frac{2557}{576*T}\right] \end{aligned}$$

IV. Disturbance Observer Design

The disturbance variable d contains the uncertainties, unmodeled quantities, and external disturbances. The equation of the link is rewritten, with the disturbance variable, as

$$I\ddot{q}_1 + Mgl\ \sin(q_1) + k(q_1 - q_2) = d \tag{16}$$

Then, the state space model (4) becomes

$$\dot{x} = f(x) + g_1(x)u + g_2(x)d \tag{17}$$

with

$$g_2(x) = [0 \quad \frac{1}{I} \quad 0 \quad 0]^T$$

The output derivatives are given in matrix form by

$$\mathrm{Y}(t) = \begin{bmatrix} y(t) \\ \dot{y}(t) \\ \ddot{y}(t) \\ \dddot{y}(t) \\ y^{(4)}(t) \end{bmatrix} = \begin{bmatrix} h \\ L_f h \\ L_f^2 h \\ L_f^3 h \\ L_f^4 h \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ L_{g_1} L_f^3 hu(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ L_{g_2} L_f hd \\ 0 \\ L_{g_2} L_f^3 hd \end{bmatrix} \tag{18}$$

where,

$$L_{g_2} L_f h = \frac{\partial L_f h}{\partial x} g_2 = \frac{1}{I}$$

$$L_{g_2} L_f^3 h = \frac{\partial L_f^3 h}{\partial x} g_2 = -\frac{Mgl}{I^2}\cos(x_1) - \frac{k}{I^2}$$

Then, the optimal control (15) has the form

$$u = -(L_{g_1} L_f^3 h)^{-1}\left\{\mathrm{KM}_4 + L_f^4 h - y_{ref}^{(4)} + L_{g_2} L_f^3 h\hat{d} + k_3 L_{g_2} L_f h\hat{d}\right\} \tag{19}$$

where,

$\hat{d}$ is the estimated disturbance.

An initial disturbance observer is given by [3]

$$\dot{\hat{d}} = -Lg_2\hat{d} + L\left(\dot{x} - f(x) - g_1 u(t)\right) \tag{20}$$

where

L is a gain vector to be designed.

$$\begin{aligned} \dot{\hat{d}} = {} & L\left(\dot{x} - f(x)\right) - Lg_2\hat{d} + Lg_1(L_{g_1} L_f^3 h)^{-1} \\ & \left\{\mathrm{KM}_4 + L_f^4 h - y_{ref}^{(4)} + L_{g_2} L_f^3 h\hat{d} + k_3 L_{g_2} L_f h\hat{d}\right\} \end{aligned} \tag{21}$$

L can be chosen as

$$L = p\left(\frac{\partial L_f^3 h}{\partial x} + k_3 \frac{\partial L_f h}{\partial x}\right) \tag{22}$$

p is a constant

The choice of the gain L will be explained in the section about the stability analysis.

The Lie derivatives used in (21) can be defined in function of L as

$$\begin{aligned} L_{g_1} L_f^3 h &= \frac{\partial L_f^3 h}{\partial x} g_1 = \frac{1}{p} Lg_1 \\ L_{g_2} L_f^3 h &= \frac{\partial L_f^3 h}{\partial x} g_2 = \frac{1}{p} Lg_2 - k_3 L_{g_2} L_f h \\ L_f^4 h &= \frac{\partial L_f^3 h}{\partial x} f = \frac{1}{p} Lf - k_3 L_f^2 h \\ L\dot{x} &= p\left(\frac{\partial L_f^3 h}{\partial x}\frac{\partial x}{\partial t} + k_3 \frac{\partial L_f h}{\partial x}\frac{\partial x}{\partial t}\right) = p\left(y^{(4)} + k_3\ddot{y}\right) \end{aligned} \tag{23}$$

Substituting (23) in (21), and after simplification, we get

$$\begin{aligned} \dot{\hat{d}} = {} & p(y^{(4)} - y_{ref}^{(4)}) + pk_4(\ddot{y} - \ddot{y}_{ref}) + pk_3(\dddot{y} - \dddot{y}_{ref}) + \\ & pk_2(\dot{y} - \dot{y}_{ref}) + pk_1(y - y_{ref}) \end{aligned} \tag{24}$$

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Then, by integrating (24), the equation of the disturbance observer is given by

$$\hat{d} = p(y^{(3)} - y_{ref}^{(3)}) + pk_4(\ddot{y} - \ddot{y}_{ref}) + pk_3(\dot{y} - \dot{y}_{ref}) + pk_2(y - y_{ref}) + pk_1\int(y - y_{ref})dt \quad (25)$$

The observer contains an integral action, which allows the elimination of the steady state error and enhances the robustness of the control scheme with respect to model uncertainties and disturbances rejection.

V. Stability Analysis

One of the main issues in control task is to guarantee the stability of the closed loop system under the derived optimal control law and the disturbance observer.

Substituting the control optimal (19) into the last equation of (18), the dynamic error of the system is given by

$$e_y^4(t)+k_4\ddot{e}_y(t)+k_3\ddot{e}_y(t)+k_2\dot{e}_y(t)+k_1e_y(t)=\left(L_{g_2}L_f^3h+k_3L_{g_2}L_fh\right)e_d(t) \quad (26)$$

where,

$e_y(t)=y(t)-y_{ref}(t)$ is the output tracking error.

$e_d(t)=d(t)-\hat{d}(t)$ is the disturbance error.

In order to guarantee the stability, the disturbance error must tend to zero. Then, it is easily to verify that the system dynamic (26) is Hurwitz.

Since, in general, there is no prior information about the derivative of the disturbance *d*, it is reasonable to suppose that

$$\dot{d} = 0 \quad (27)$$

this implies that the disturbance varies slowly relative to the observer dynamics [4].

Then, from (17) and (20), the error dynamic of the disturbance observer is given by

$$\dot{e}_d(t) + Lg_2e_d(t) = 0 \quad (28)$$

It can be shown that the observer is exponentially stable by choosing

$$Lg_2 > 0 \quad (29)$$

Then,

$$\lim_{t\to\infty} e_d(t) = 0 \quad (30)$$

In general, it is not easy to select the nonlinear function *L*. However, for a single-link flexible joint robot, with the help of its characteristics and from its dynamic error (26), the function *L* is chosen as shown above in (22), such that the observer given by (20) is asymptotically stable.

From the definition (22), the choice of *p* depends of robot's characteristics and prediction time. It is taken with respect of the condition

$$p\left(-\frac{Mgl}{I^2}\cos(x_1) - \frac{k}{I^2} + \frac{k_3}{I}\right) > 0 \quad (31)$$

VI. Simulation Results

In this section, simulations are conducted to test the performance of the proposed control strategy. The robot parameters are M=0.2 kg, L=0.02m, I=1.35×10^{-4} kg-m^2, k=7.47 N-m/rad, and J=2.16×10^{-3} kg-m^2. The controller parameters are chosen by trial and error, with respect to condition (31), in order to get an acceptable tracking.

First, the robot is controlled by the predictive control law without disturbance observer. The information about the disturbance (Fig. 2) is not included in the controller computation. The system is run with the nominal values of the robot parameters. The tracking performance of the angular displacement to a smooth step reference is shown in Fig. 3. The result shows an acceptable performance. However, a steady error occurs with the disturbance variations. Then, the disturbance observer is taken into account while carrying out the control law. The tracking performance in case of a smooth step reference and sinusoidal reference are shown in Fig. 4 and Fig. 5 respectively. Fig. 6 and Fig. 7 give the estimation of the disturbance for both references respectively. As shown in results, the tracking performance is achieved successfully and the effect of disturbance is well rejected.

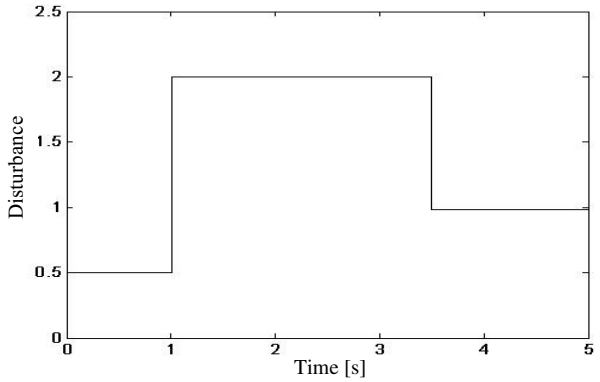


Fig. 2 External disturbance applied to the robot

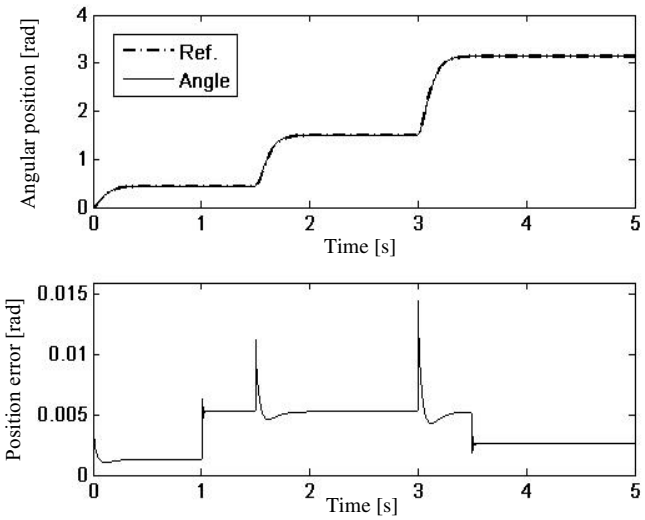


Fig. 3 a. Angular position of the link with smooth step reference under NPC controller without disturbance observation.
b. Tracking error.

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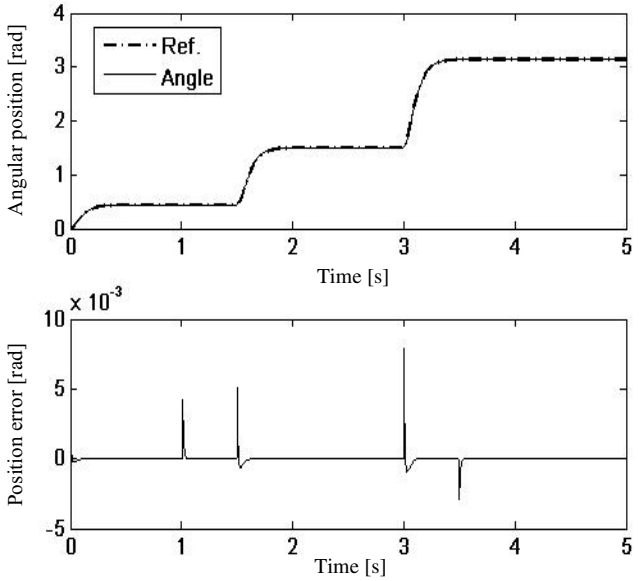


Fig. 4 a. Angular position of the link with smooth step reference
b. Tracking error (T= 10^{-3}s and p=10^{-9})

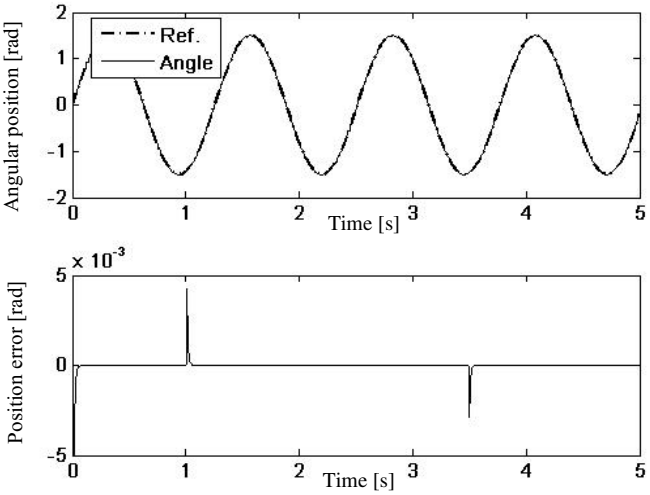


Fig. 5 a. Angular position of the link with sinusoidal reference
b. Tracking error (T= 10^{-3}s and p=10^{-9})

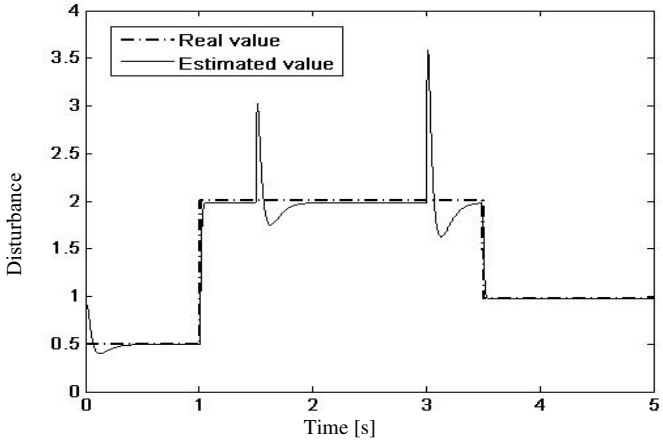


Fig. 6 External disturbance estimation for smooth step reference

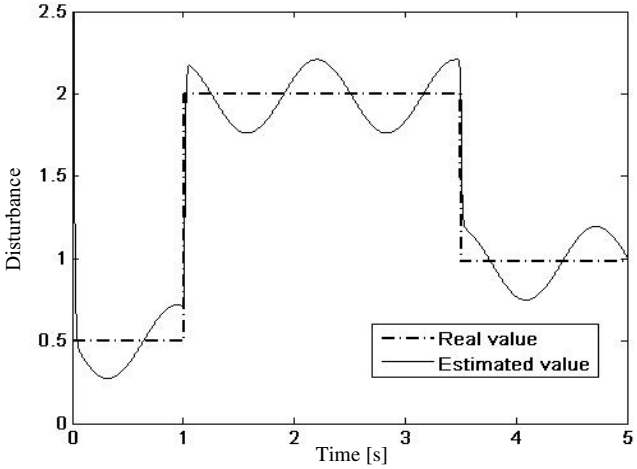


Fig. 7 External disturbance estimation for sinusoidal reference

Then, in case of mismatched model, the robot parameters are varied 50 %. These variations are not taken into account when the control law is carried out. The nominal values of parameters are used in the control law computation. The same values of prediction time and observer gain p, as above, are used in this simulation.

The tracking performances for angular displacement with smooth step and sinusoidal references are shown in Fig. 8 and Fig. 9 respectively. These results show that the robustness to parameters variations and external disturbance is successfully achieved by this control strategy. The disturbance observer takes care about the unmodeled quantities, uncertainties and unknown external disturbances.

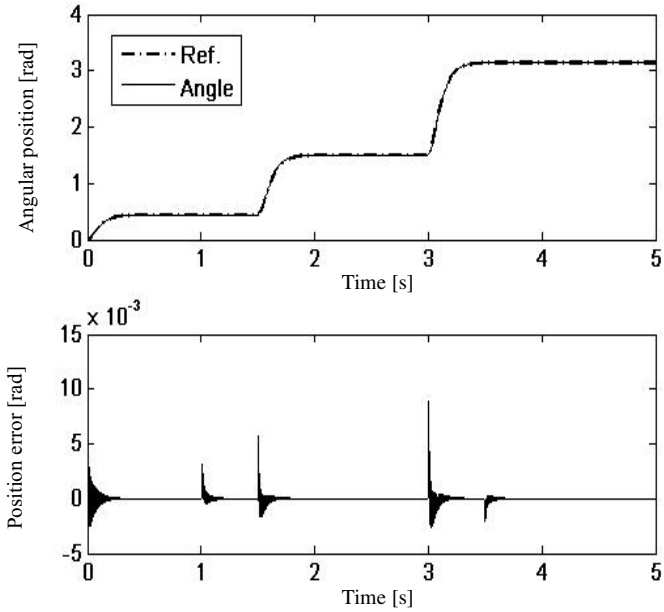


Fig. 8 a. Angular position of the link with smooth step reference in case of mismatched model
b. Tracking error

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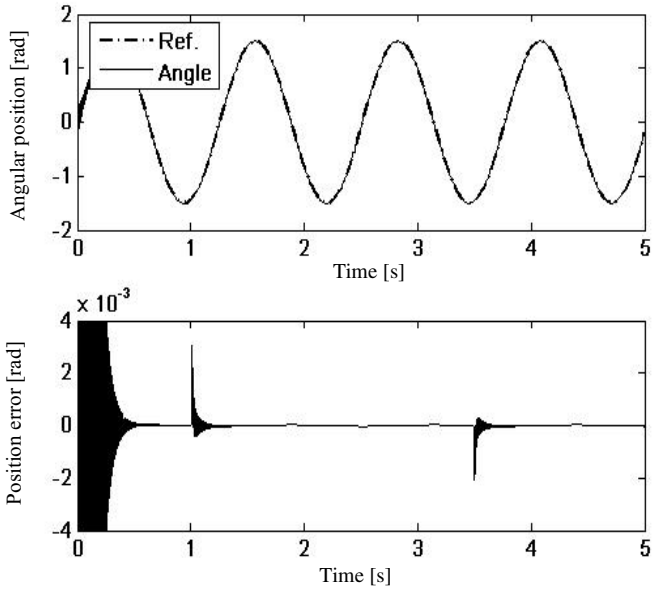


Fig. 9 a. Angular position of the link with sinusoidal reference in case of mismatched model
b. Tracking error

V. CONCLUSIONS

In this paper, a nonlinear predictive control method with a disturbance observer is applied to single-link flexible joint robot. The controller is robust with respect to modeling errors, very effective in disturbance rejection, and gives no steady state error caused by either parameters uncertainties or external disturbance.

The asymptotic stability of the closed loop system is guaranteed. The controller parameters are easy to choose under certain conditions. Results show that the single-link flexible-joint robot under nonlinear predictive control has good output tracking performance.

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